

Lesson 03 Demo 04

Enabling MFA for IAM User

Objective: To enable Multi-Factor Authentication (MFA) for an IAM user, enhancing the security of AWS accounts

Tools required: AWS Console and Google Authenticator

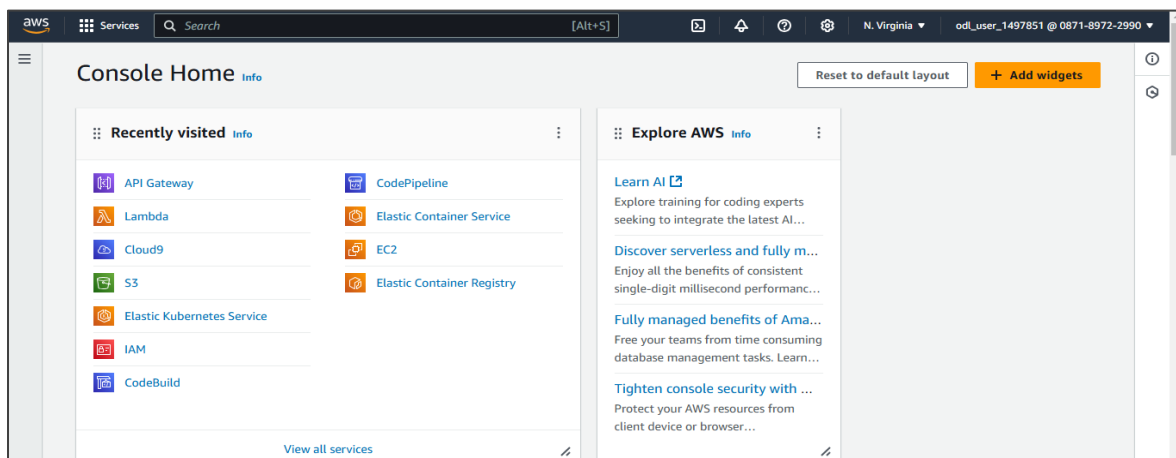
Prerequisites: None

Steps to be followed:

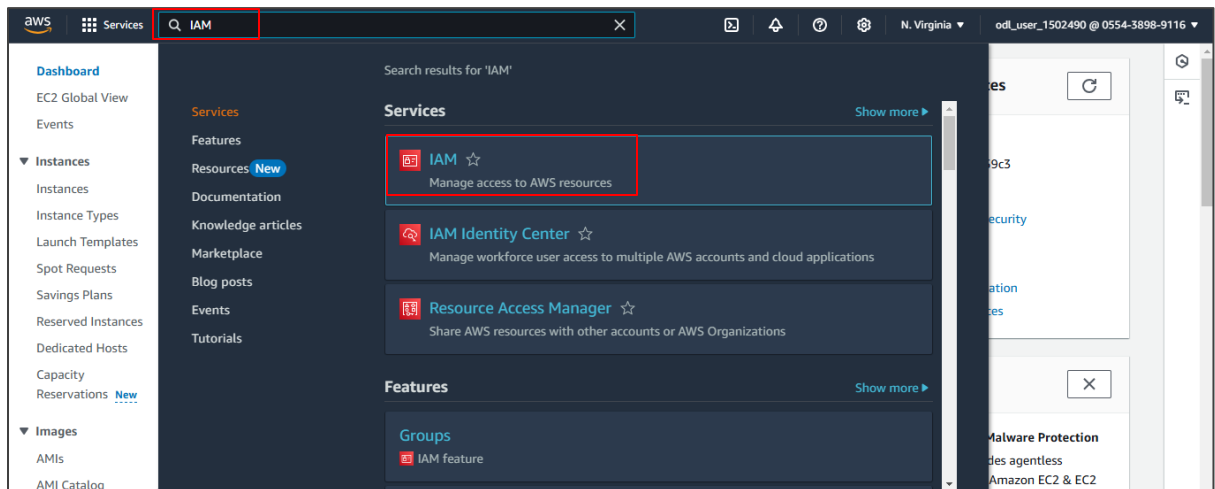
1. Create an IAM user
2. Install Google Authenticator
3. Assign an MFA device to the IAM User
4. Sign in with MFA

Step 1: Create an IAM user

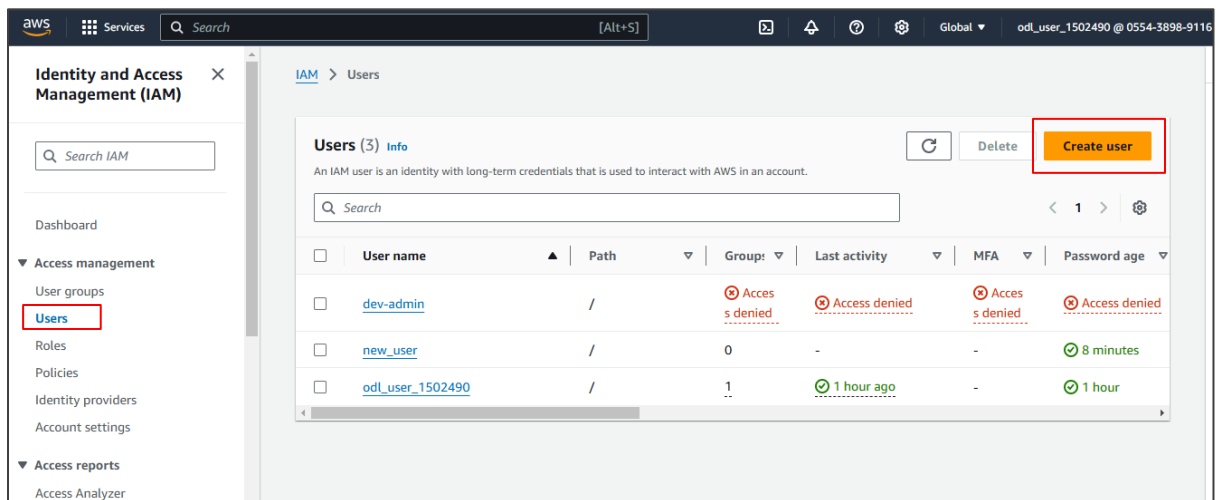
1.1 Open the **AWS Management Console** in your practice environment



1.2 Search for IAM in the search bar and click on the IAM service



1.3 Choose Users in the left navigation pane and click on Create user



1.4 Enter a desired name in the **User name** field

The screenshot shows the 'Specify user details' step in the AWS IAM console. The left sidebar lists the steps: Step 1 (Specify user details), Step 2 (Set permissions), Step 3 (Review and create), and Step 4 (Retrieve password). The main content area is titled 'Specify user details' and contains a 'User details' section. In this section, the 'User name' field is highlighted with a red box and contains the text 'Demo_User'. Below the field, a note states: 'The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)'. There is an unchecked checkbox for 'Provide user access to the AWS Management Console - optional' with a note: 'If you're providing console access to a person, it's a [best practice](#) to manage their access in IAM Identity Center.' At the bottom of the 'User details' section, there is a blue information box with a note: 'If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)'. At the bottom right of the console, there are 'Cancel' and 'Next' buttons.

1.5 Select the check box of AWS Management Console access, choose **Custom password**, and set a password

The screenshot shows the 'Set permissions' step in the AWS IAM console. The left sidebar lists the steps: Step 1 (Specify user details), Step 2 (Set permissions), Step 3 (Review and create), Step 4 (Retrieve password), and Step 5 (Set permissions). The main content area is titled 'Set permissions' and contains a 'User details' section. In this section, the 'User name' field is highlighted with a red box and contains the text 'Demo_User'. Below the field, a note states: 'The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)'. There is a checked checkbox for 'Provide user access to the AWS Management Console - optional' with a note: 'If you're providing console access to a person, it's a [best practice](#) to manage their access in IAM Identity Center.' Below this, there is a 'Console password' section with two options: 'Autogenerated password' (unchecked) and 'Custom password' (checked). The 'Custom password' option is highlighted with a red box. Below the 'Custom password' option, there is a text input field for the password, which is currently empty. Below the input field, there are two bullet points: 'Must be at least 8 characters long' and 'Must include at least three of the following mix of character types: uppercase letters (A-Z), lowercase letters (a-z), numbers (0-9), and symbols ! @ # \$ % ^ & * () _ + - (hyphen) = [] { } | ' '. Below the input field, there is an unchecked checkbox for 'Show password'. At the bottom of the 'User details' section, there is a checked checkbox for 'Users must create a new password at next sign-in - Recommended' with a note: 'Users automatically get the [IAMUserChangePassword](#) policy to allow them to change their own password.' At the bottom right of the console, there are 'Cancel' and 'Next' buttons.

1.6 Deselect **User must create a new password at next sign-in - Recommended** and click on **Next**

Retrieve password

☒ Provide user access to the AWS Management Console - optional
If you're providing console access to a person, it's a [best practice](#) to manage their access in IAM Identity Center.

Console password

☐ Autogenerated password
You can view the password after you create the user.

☒ Custom password
Enter a custom password for the user.

- Must be at least 8 characters long
- Must include at least three of the following mix of character types: uppercase letters (A-Z), lowercase letters (a-z), numbers (0-9), and symbols ! @ # \$ % ^ & * () _ + - (hyphen) = [] { } | ' "

☐ Show password

☐ **User must create a new password at next sign-in - Recommended**
Users automatically get the [IAMUserChangePassword](#) policy to allow them to change their own password.

① If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)

Cancel **Next**

1.7 Keep the default settings and click **Next**

Step 4
Retrieve password

Set permissions boundary - optional

User groups (2)

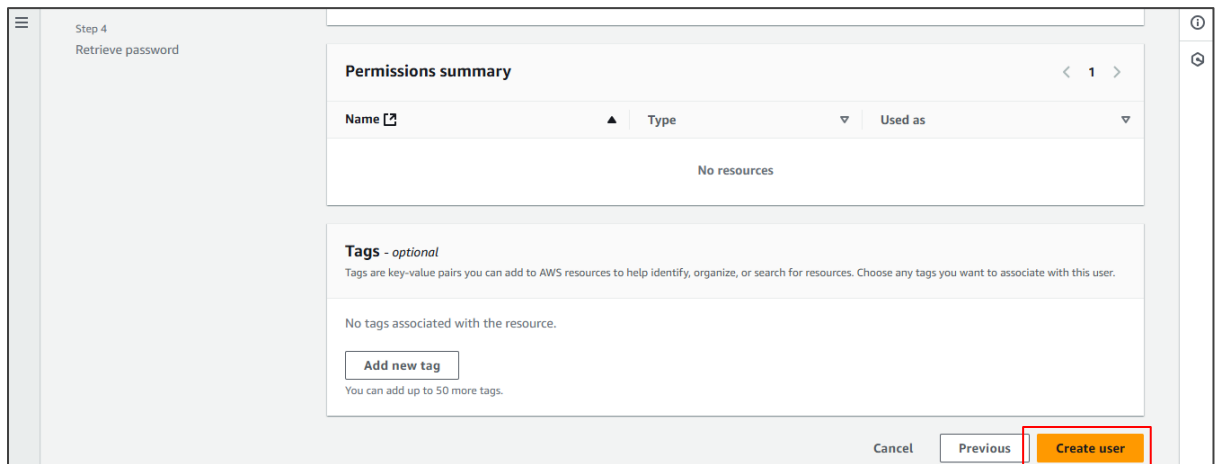
Search

☐	Group name	Users	Attached policies	Created
☐	Admins	0	-	2022-01-20 (2 years ...)
☐	Attendees	1	AWS14910-PolicyDefinit...	2024-11-03 (1 hour a ...)

▶ Set permissions boundary - optional

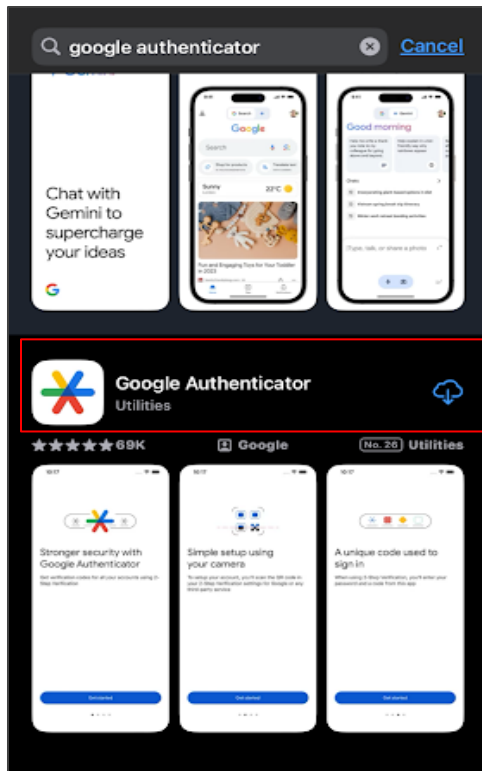
Cancel Previous **Next**

1.8 Click on **Create user**



Step 2: Install Google Authenticator

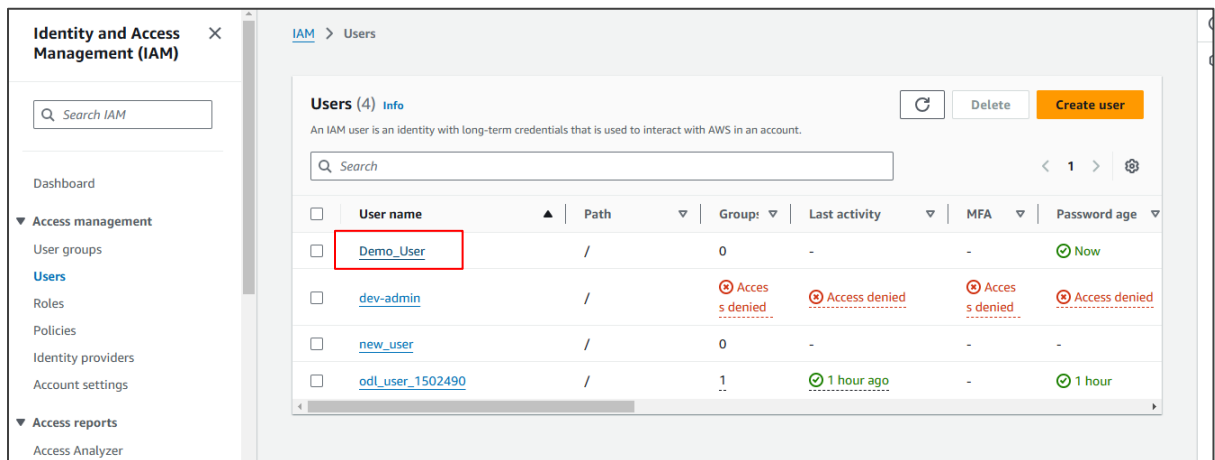
2.1 Install Google authenticator on your mobile device



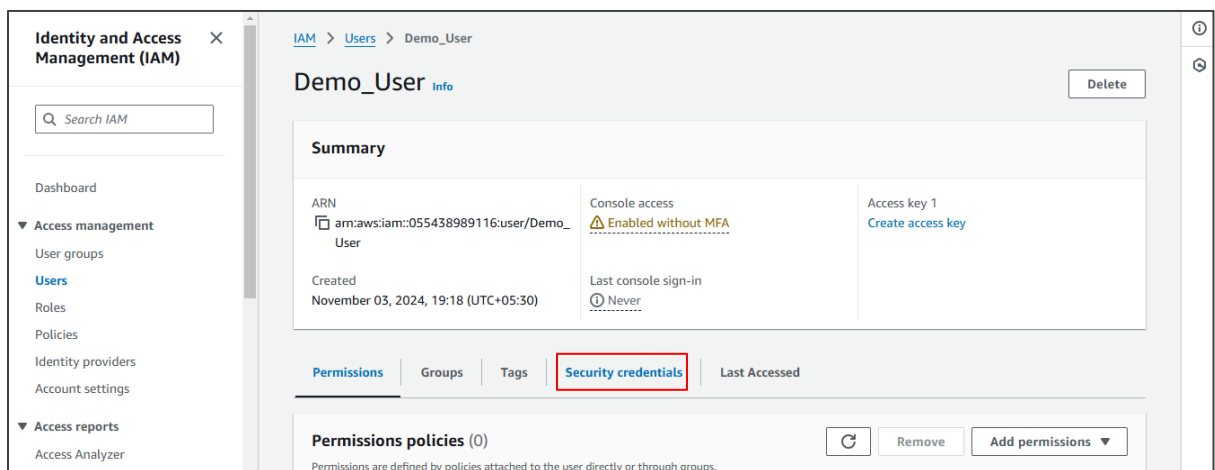
Open the app and get ready for the next steps.

Step 3: Assign an MFA device to the IAM User

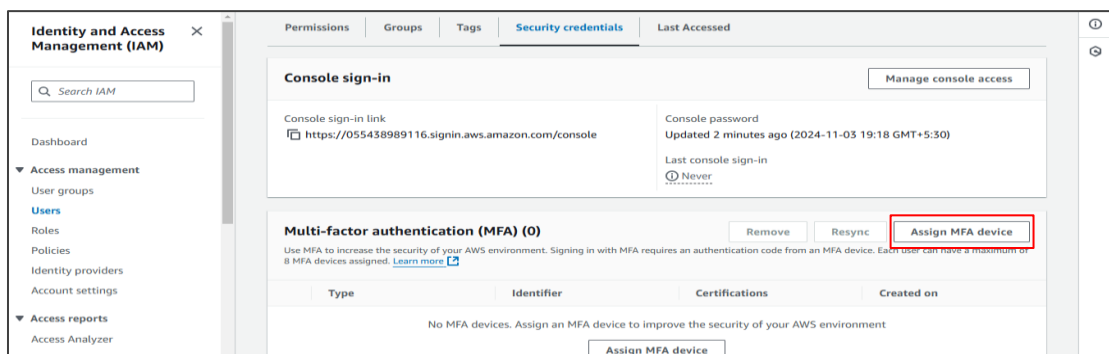
3.1 Navigate back to the AWS Management Console and click on the created IAM user



3.2 Click on Security credentials



3.3 Click on Assign MFA device



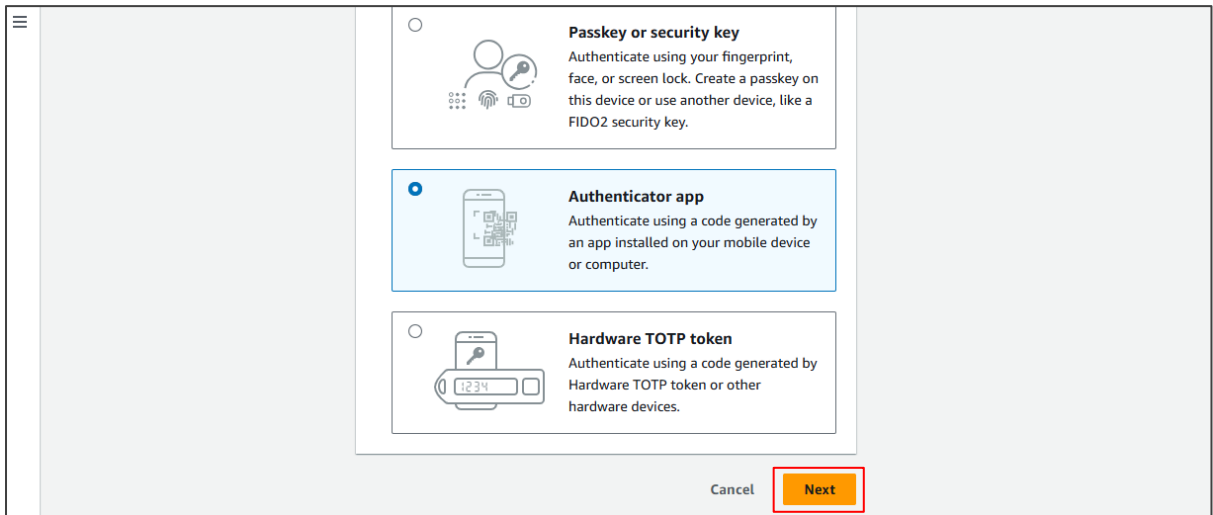
3.4 Provide a name for the MFA device in the **Device name** field

The screenshot shows the AWS IAM console interface for assigning an MFA device to a user. The breadcrumb navigation at the top reads: IAM > Users > Demo_User > Assign MFA device. On the left, a sidebar indicates the current step is 'Step 1: Select MFA device', with 'Step 2: Set up device' listed below it. The main content area is titled 'Select MFA device' with an 'Info' link. It contains two sections: 'MFA device name' and 'MFA device'. The 'MFA device name' section has a text input field labeled 'Device name' with the placeholder text 'This name will be used within the identifying ARN for this device.' The field contains the text 'demo_device' and is highlighted with a red rectangular box. Below the field is a note: 'Maximum 64 characters. Use alphanumeric and '+', '=', '@', '-', '_' characters.' The 'MFA device' section is titled 'MFA device' and contains a 'Device options' subsection with the text: 'In addition to username and password, you will use this device to authenticate into your account.' Below this, there are two selectable options: 'Passkey or security key' (which is currently selected, indicated by a blue circle) and 'Authenticator app' (which is not selected).

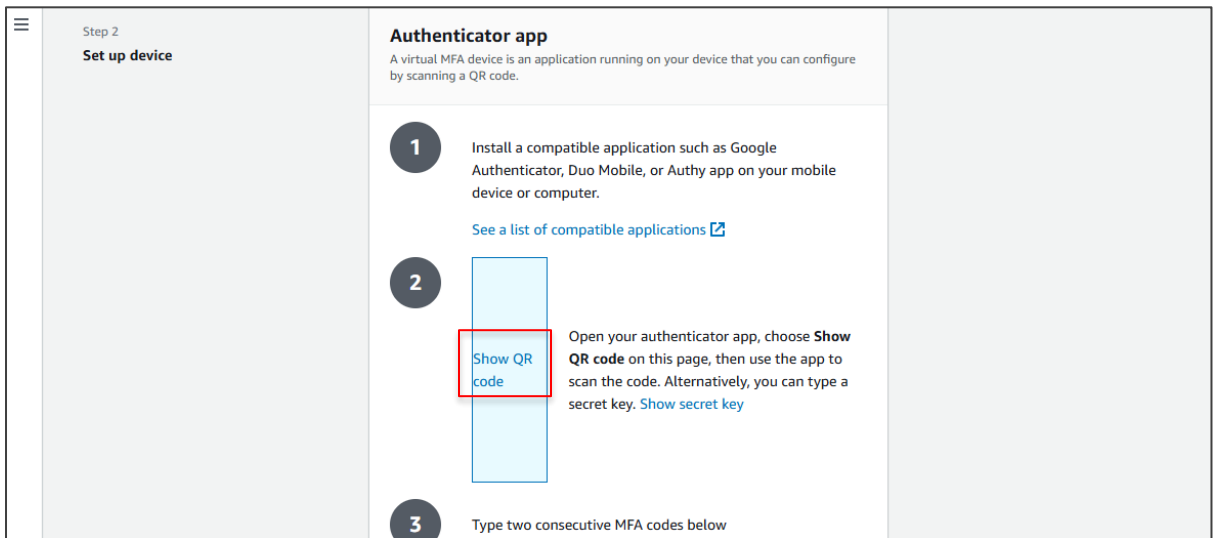
3.5 Select **Authenticator app**

This screenshot shows the same 'Assign MFA device' page, but with the 'Authenticator app' option selected. The 'Passkey or security key' option is now unselected. The 'Authenticator app' option is highlighted with a red rectangular box. It features an icon of a smartphone displaying a QR code. The text for this option reads: 'Authenticator app' followed by 'Authenticate using a code generated by an app installed on your mobile device or computer.' The 'Passkey or security key' option remains visible above it, with its own icon and description: 'Authenticate using your fingerprint, face, or screen lock. Create a passkey on this device or use another device, like a FIDO2 security key.'

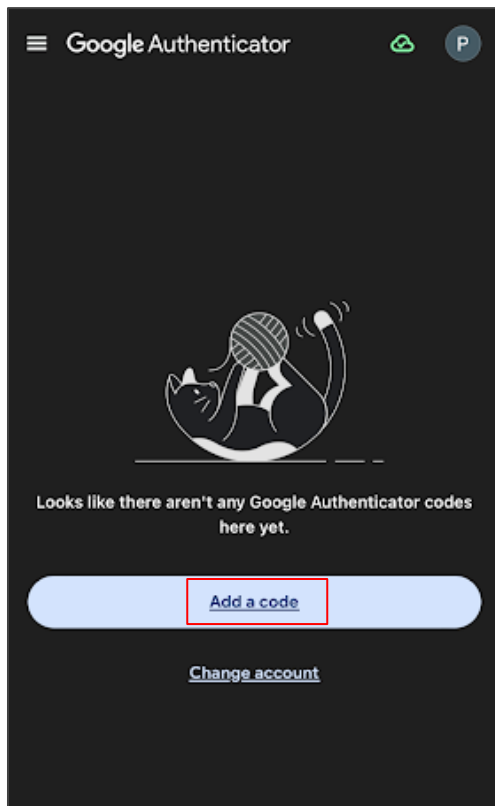
3.6 Click on **Next**



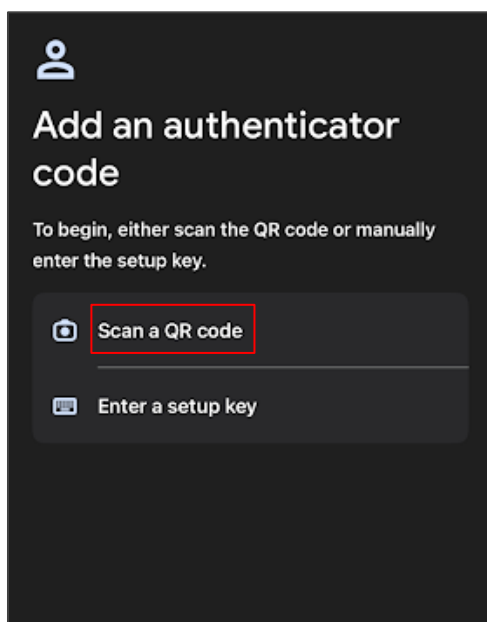
3.7 Click on **Show QR code**



3.8 Open **Google Authenticator** on your mobile device and tap on **Add a code**



3.9 Select **Scan a QR code**



3.10 Scan the QR code displayed on the AWS Management Console with your mobile device

The screenshot shows the AWS Management Console interface. At the top, there's a navigation bar with the AWS logo, 'Services', a search bar, and user information 'odl_user_1502490 @ 0554-3898-9116'. The main content area displays a multi-step setup process. Step 2, indicated by a circled '2', shows a QR code. To the right of the QR code, text instructs the user to open their authenticator app, choose 'Show QR code', and scan the code. It also mentions an alternative of typing a secret key with a link 'Show secret key'. Step 3, indicated by a circled '3', is titled 'Type two consecutive MFA codes below'. It contains two input fields. The first field is labeled 'Enter a code from your virtual app below' and the second is labeled 'Wait 30 seconds, and enter a second code entry.'. At the bottom of the form, there are three buttons: 'Cancel', 'Previous', and 'Add MFA'.

3.11 Enter the generated MFA codes from your device and click on **Add MFA**

This screenshot shows the same AWS Management Console MFA setup screen as the previous one, but now with data entered into the form. In step 3, the first input field contains the code '285722' and the second input field contains '772478'. Both input fields are highlighted with red rectangles. The 'Add MFA' button at the bottom right is also highlighted with a red rectangle. The rest of the interface, including the navigation bar and step 2 with the QR code, remains the same.

3.12 Confirm that the MFA device has been successfully assigned

The screenshot shows the AWS IAM console interface. On the left is a navigation sidebar with 'Identity and Access Management (IAM)' selected. The main content area displays the 'Demo_User' profile. A green banner at the top states 'MFA device assigned'. The 'Summary' section shows the user's ARN, creation date, and console access status. The 'Console access' status is 'Enabled with MFA', which is highlighted with a red box. Other details include 'Access key 1' and 'Last console sign-in'.

Identity and Access Management (IAM)

Search IAM

Dashboard

Access management

- User groups
- Users**
- Roles
- Policies
- Identity providers
- Account settings

Access reports

- Access Analyzer

MFA device assigned

You can register up to 8 MFA devices of any combination of the currently supported MFA types with your AWS account root and IAM user. With multiple MFA devices, you only need one MFA device to sign in to the AWS console or create a session through the AWS CLI with that user.

IAM > Users > Demo_User

Demo_User Info Delete

Summary

ARN arn:aws:iam::055438989116:user/Demo_User	Console access Enabled with MFA	Access key 1 Create access key
Created November 03, 2024, 19:18 (UTC+05:30)	Last console sign-in Never	

Permissions Groups Tags **Security credentials** Last Accessed

Step 4: Sign in with MFA

4.1 Log in using the IAM user credentials created and click on **Sign in**

The screenshot shows the AWS IAM user sign-in page. The 'IAM user sign in' section contains fields for 'Account ID (12 digits) or account alias' (055438989116), 'IAM username' (Demo_User), and 'Password'. The 'IAM username' and 'Password' fields are highlighted with a red box. Below these fields are checkboxes for 'Show Password' and 'Remember this account', a 'Sign in' button, and links for 'Having trouble?' and 'Create a new AWS account'. To the right is an 'Amazon Lightsail' advertisement featuring a robot character.

IAM user sign in

Account ID (12 digits) or account alias
055438989116

IAM username
Demo_User

Password

☐ Show Password [Having trouble?](#)

Sign in

[Sign in using root user email](#)

[Create a new AWS account](#)

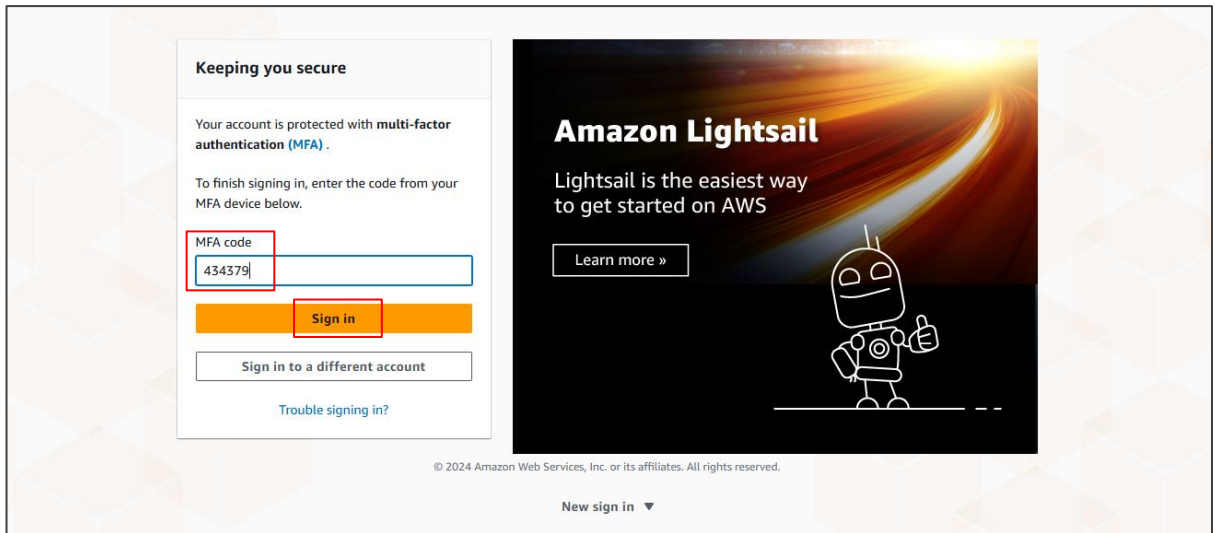
☐ Remember this account

Amazon Lightsail

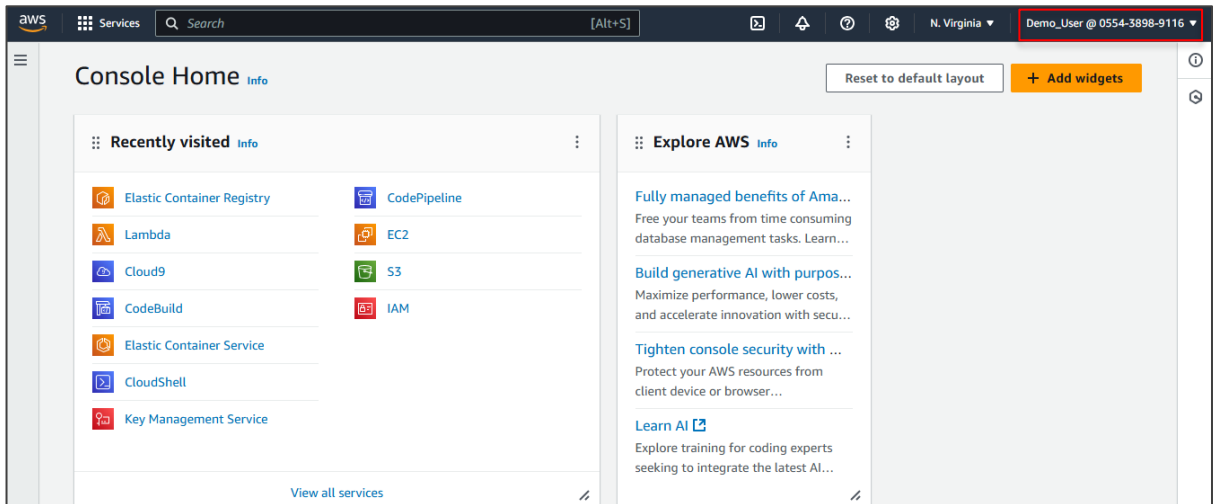
Lightsail is the easiest way to get started on AWS

[Learn more »](#)

4.2 Enter the MFA code generated by your mobile device and click on **Sign in**



4.3 Confirm that you have successfully signed in to the IAM user account with MFA enabled



By following these steps, you have successfully enabled Multi-Factor Authentication (MFA) for an IAM user to enhance the security of AWS accounts.